

✓ Exploratory Data Analysis (EDA) Project

Objective:

To analyze customer sales data, identify patterns, find relationships between variables, and detect data quality issues using Python visualization.

Dataset:

Customer Sales Dataset

Tools Used:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

```
from google.colab import files
```

```
uploaded = files.upload()
```

customer_sales_dataset.csv.xlsx

customer_sales_dataset.csv.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 11623 bytes, last modified: 19/06/2026 - 100% done
Saving customer_sales_dataset.csv.xlsx to customer_sales_dataset.csv.xlsx

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
df = pd.read_excel("customer_sales_dataset.csv.xlsx")
```

```
df.head()
```

	Customer_ID	Age	Gender	City	Product	Quantity	Price	Discount	S
0	101	25	Male	Vijayawada	Laptop	1	55000	5	5
1	102	32	Female	Hyderabad	Mobile	2	25000	10	4
2	103	28	Male	Chennai	Tablet	1	18000	0	1
3	104	45	Female	Bangalore	Laptop	1	65000	8	5
4	105	22	Male	Vijayawada	Mobile	3	20000	5	5

Next steps:

[Generate code with df](#)[New interactive sheet](#)

df.shape

(30, 10)

df.columns

```
Index(['Customer_ID', 'Age', 'Gender', 'City', 'Product', 'Quantity',
      'Price',
      'Discount', 'Sales', 'Rating'],
      dtype='object')
```

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Customer_ID     30 non-null    int64
1   Age             30 non-null    int64
2   Gender          30 non-null    object
3   City            30 non-null    object
4   Product         30 non-null    object
5   Quantity        30 non-null    int64
6   Price           30 non-null    int64
7   Discount        30 non-null    int64
8   Sales           30 non-null    int64
9   Rating          30 non-null    float64
dtypes: float64(1), int64(6), object(3)
memory usage: 2.5+ KB
```

df.describe()

	Customer_ID	Age	Quantity	Price	Discount	Sales
count	30.000000	30.000000	30.000000	30.000000	30.000000	30.000000
mean	115.500000	33.733333	1.633333	37116.666667	8.433333	45172.000000
std	8.803408	9.066548	0.808717	26777.397614	6.333727	26574.32458
min	101.000000	19.000000	1.000000	4500.000000	0.000000	9000.000000
25%	108.250000	27.250000	1.000000	18500.000000	5.000000	20500.000000
50%	115.500000	32.500000	1.000000	25000.000000	8.000000	46250.000000
75%	122.750000	39.750000	2.000000	63750.000000	10.000000	63270.000000
max	130.000000	52.000000	4.000000	90000.000000	25.000000	110500.000000

```
df.isnull().sum()
```

	0
Customer_ID	0
Age	0
Gender	0
City	0
Product	0
Quantity	0
Price	0
Discount	0
Sales	0
Rating	0

dtype: int64

```
df.duplicated().sum()
```

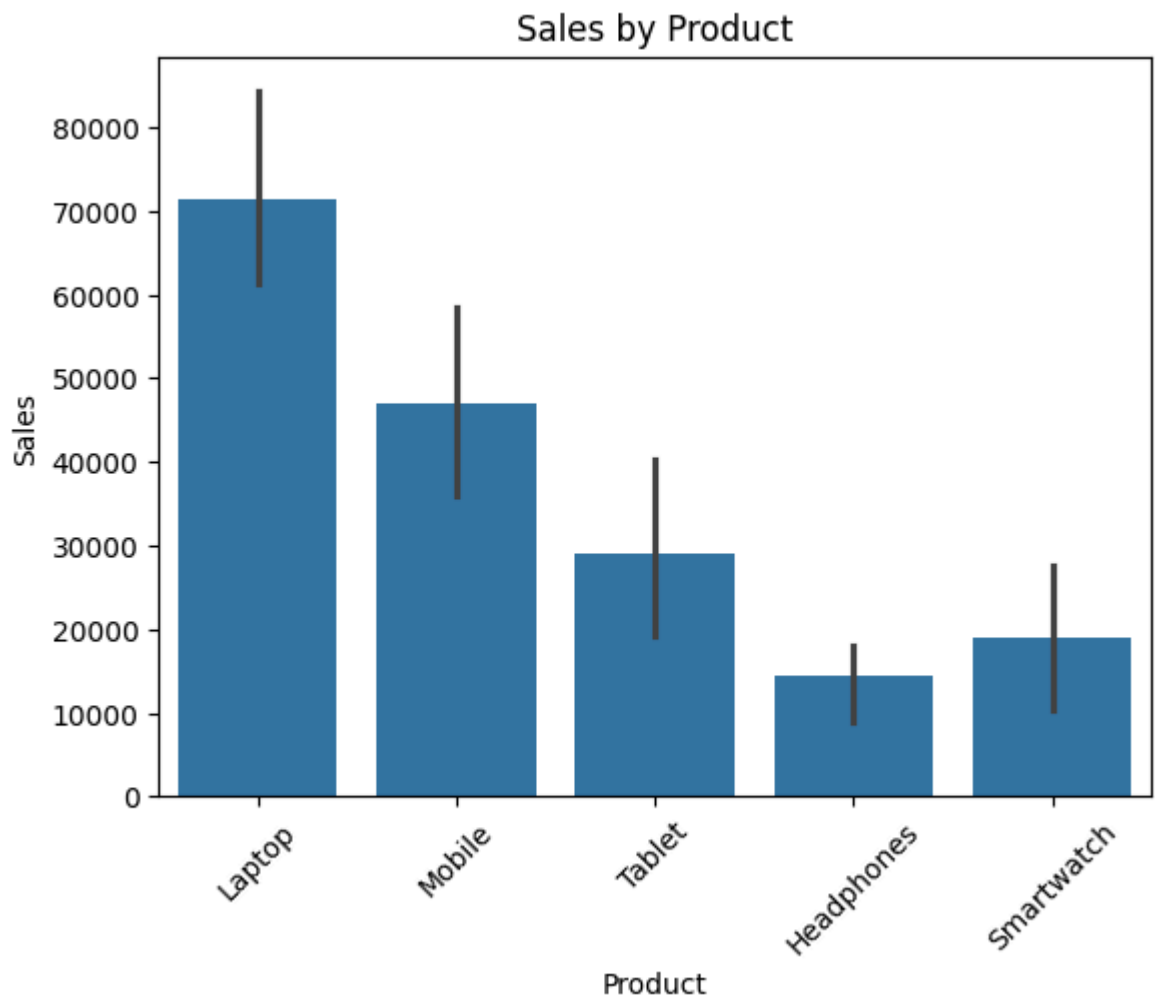
```
np.int64(0)
```

EDA Questions:

1. Which product has the highest sales?
2. Which city generates maximum revenue?
3. Does discount affect sales?
4. What is the average customer age?
5. Are there any outliers in sales?
6. Is there a relationship between rating and sales?

```
sns.barplot(x="Product", y="Sales", data=df)
```

```
plt.title("Sales by Product")  
plt.xticks(rotation=45)  
plt.show()
```



```
product_sales = df.groupby("Product")["Sales"].sum()
```

```
product_sales
```

Sales	
Product	
Headphones	43200
Laptop	714410
Mobile	376500
Smartwatch	76000
Tablet	145050

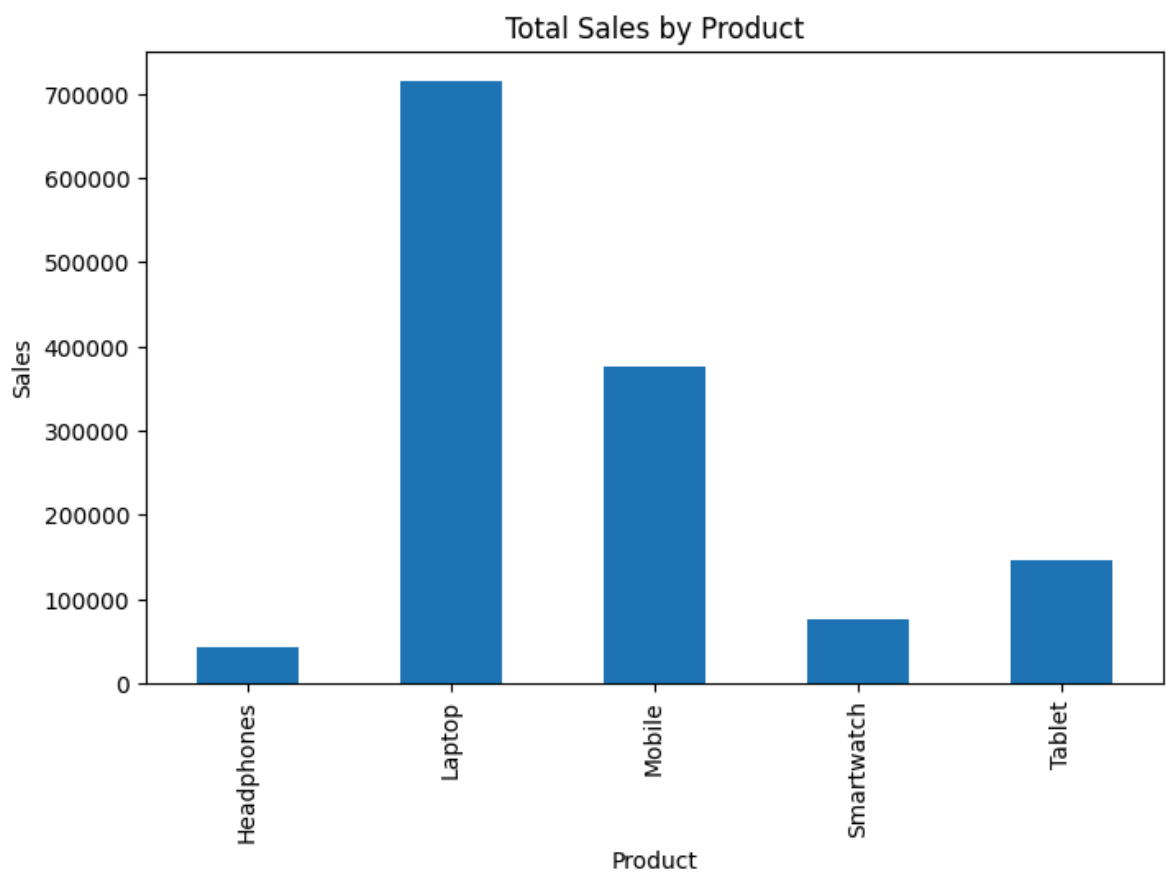
dtype: int64

```
plt.figure(figsize=(8,5))

product_sales.plot(kind="bar")

plt.title("Total Sales by Product")
plt.xlabel("Product")
plt.ylabel("Sales")

plt.show()
```



```
city_sales = df.groupby("City")["Sales"].sum()
```

```
city_sales
```

	Sales
City	
Bangalore	278760
Chennai	160600
Delhi	171600
Hyderabad	306100
Mumbai	126850
Vijayawada	311250

```
dtype: int64
```

```
plt.figure(figsize=(8,5))
```

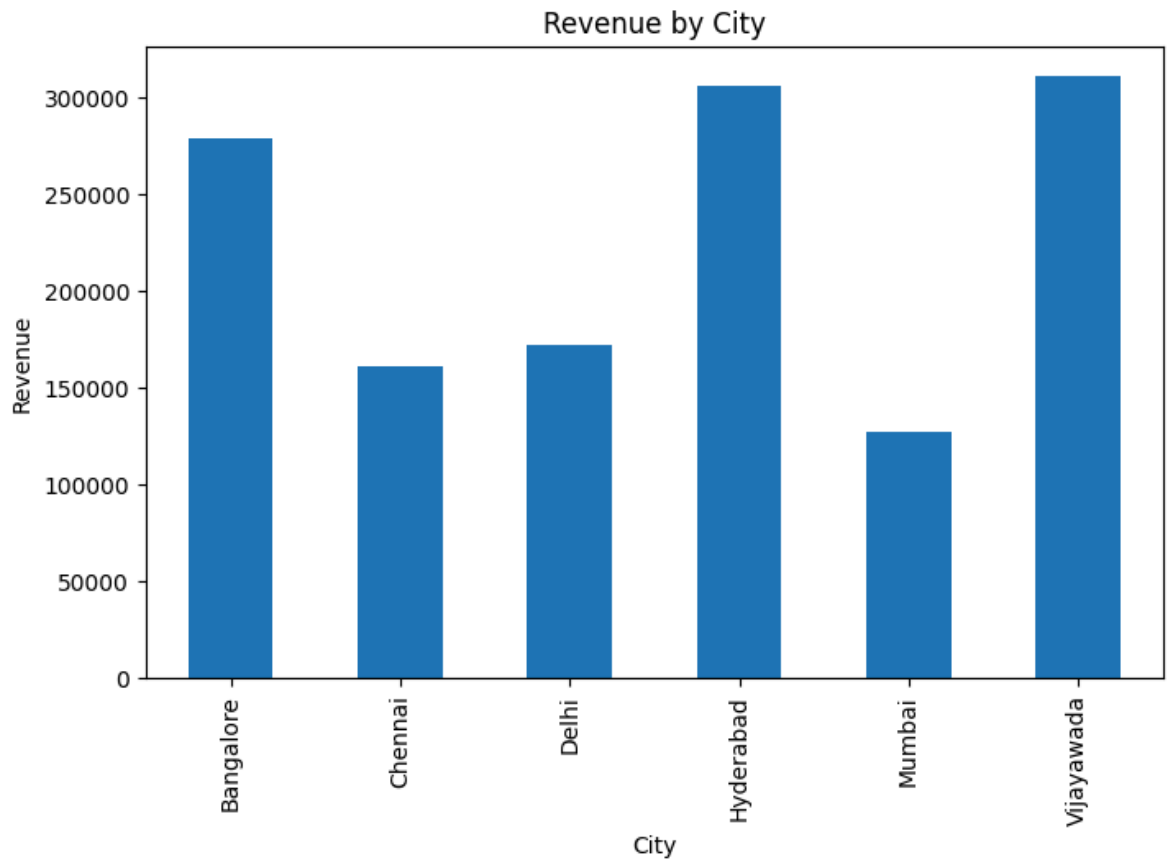
```
city_sales.plot(kind="bar")
```

```
plt.title("Revenue by City")
```

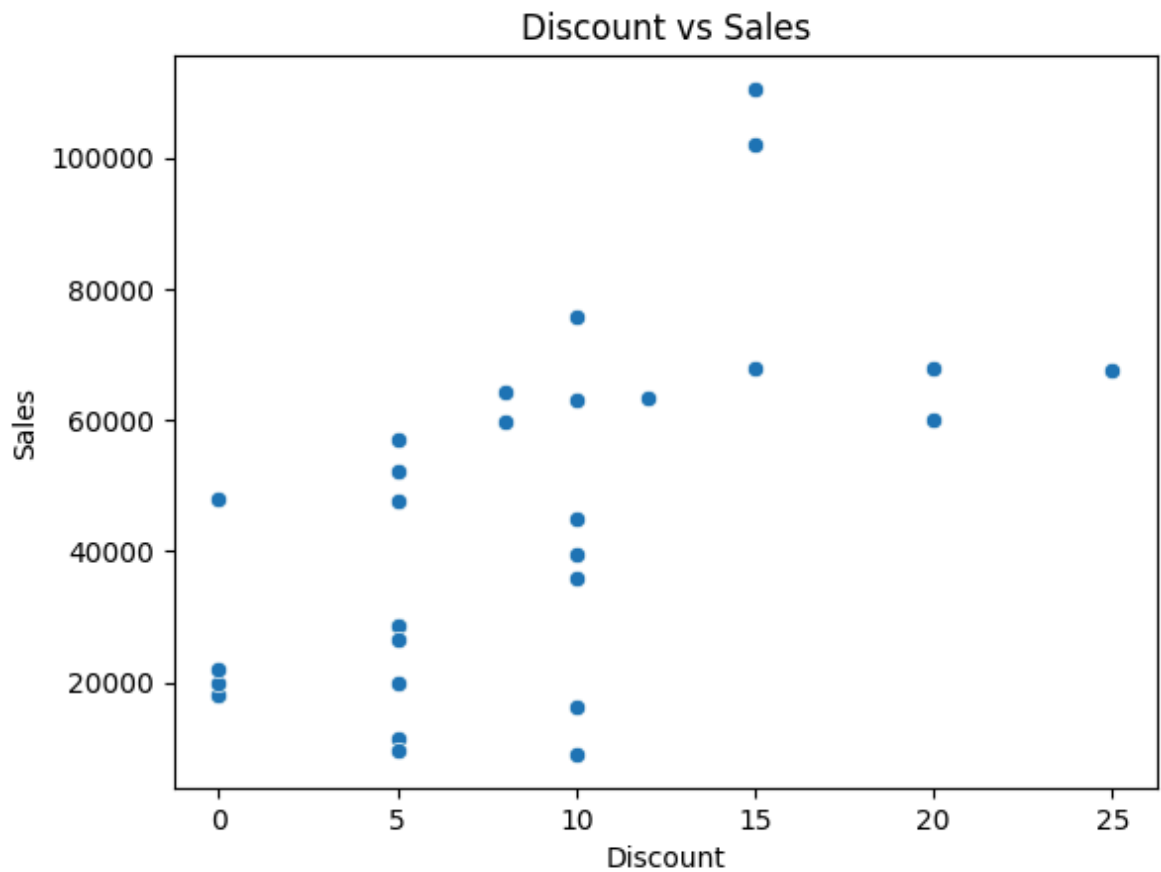
```
plt.xlabel("City")
```

```
plt.ylabel("Revenue")
```

```
plt.show()
```



```
sns.scatterplot(  
    x="Discount",  
    y="Sales",  
    data=df  
)  
  
plt.title("Discount vs Sales")  
  
plt.show()
```



```
average_age = df["Age"].mean()
```

```
average_age
```

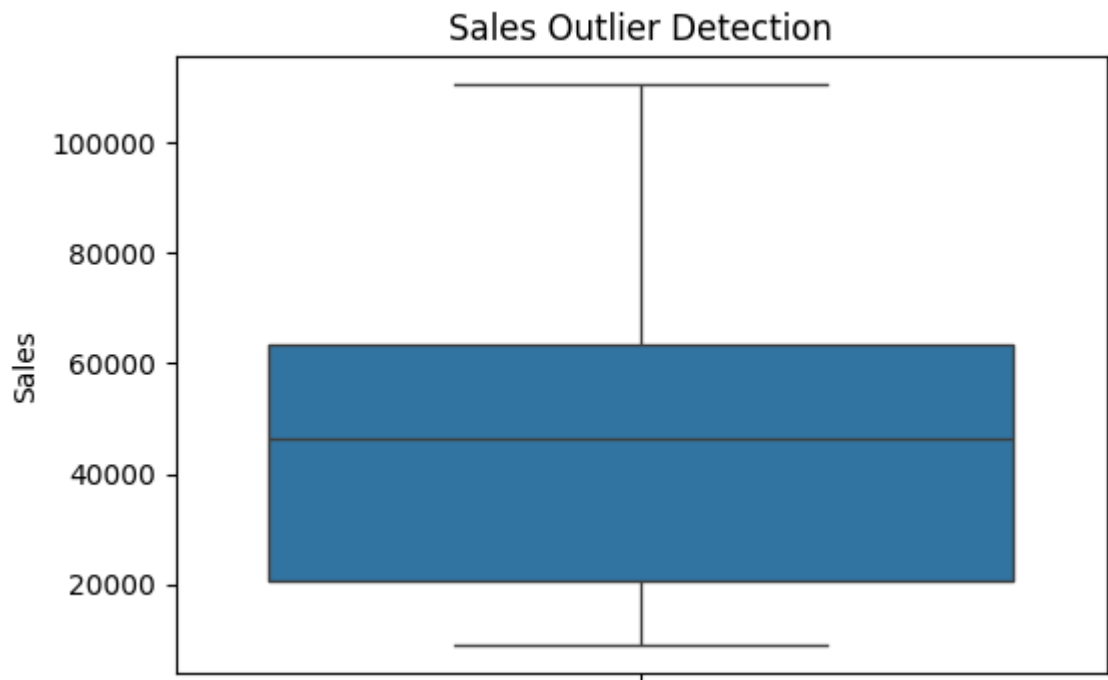
```
np.float64(33.733333333333334)
```

```
plt.figure(figsize=(6,4))
```

```
sns.boxplot(  
    y=df["Sales"]  
)
```

```
plt.title("Sales Outlier Detection")
```

```
plt.show()
```

```
sns.scatterplot(  
    x="Rating",  
    y="Sales",  
    data=df  
)  
  
plt.title("Rating vs Sales")  
  
plt.show()
```

```
plt.figure(figsize=(8,5))

sns.heatmap(
    df.corr(numeric_only=True),
    annot=True
)

plt.title("Correlation Heatmap")

plt.show()
```

